1. Introduction and Background

Pain is one of the most important signals doctors use to diagnose patients. Pain is the brain's way of alerting someone to a problem within the body and has been described as the fifth vital sign (Gouverneur et al., 2024). However, measuring pain quantitatively is challenging. Pain scales such as Numerical Rating Scales and Visual Analogue Scales, which rely on self-reported, subjective data from patients, are the current gold standard for measuring the fifth vital sign (Gouverneur et al., 2024). This system has many flaws; it can vary for every patient, is subjective, and makes it difficult to take measurements from children, people with intellectual disabilities, and those who struggle with nonverbal communication. For these reasons, it is desirable to create a new system for measuring pain that relies on objective data collected from patients. Pain can elicit responses in various bio signals, and with the advent of machine learning, these responses can be used to classify pain objectively, so doctors can better understand how much pain a patient is experiencing. It may even be possible to create a wearable device that tracks pain continuously, giving doctors a clearer view of a patient's pain over time.

Pain perception is a complex neurophysiological process that varies greatly between individuals and does not scale linearly with stimulus intensity. Current clinical tools like the Visual Analog Scale rely on conscious, subjective patient input, making them unreliable across all patient populations. The autonomic nervous system, however, responds to pain in objectively measurable ways, including shifts in heart rate variability (HRV), electrodermal activity, respiration, and peripheral vasoconstriction. Prior work has confirmed this: Kasaeyan Naeini et al. (2021), Susam et al. (2018), and Campbell et al. (2019) each demonstrated that ECG, EDA, and EMG features can reliably capture pain responses. Most of these studies, however, frame pain detection as binary and struggle to generalize across subjects. This is what our continuous regression-based approach directly targets.

2. Motivation and goals

The main motivation behind pain perception regression is to be able to predict a continuous pain score given physiological data to improve the ability to monitor and assess pain in an objective way in clinical and real-life settings. The team is especially interested in pain perception because of the inability of current wearable devices, that we possess, to address pain perception in an everyday setting. Many datasets incorporate a wide range of biosignals which are not practical or currently implementable as seen with wearable health devices. This limitation is represented in another motivation/goal of the study: compare the ability of a multi-signal dataset to a single signal measurement device in looking at pain perception. The goal of the project is to develop regression models that estimate the perceived pain levels from the physiological signals and address how different signals in conjunction can predict pain perception. One model with multiple signals will be trained as a scenario with full physiological information while a second model will incorporate only signals from wearable devices. Through comparing the two models, the goal to determine if wearable devices can quantify pain estimation is achieved.

3. Methodology

The PainMonit dataset contains two subsets: an experimental portion with controlled pain stimuli, and a clinical portion with categorical labels. This project focuses on the experimental subset for

regression modeling. Recordings are synchronized across eight channels at 250 Hz, capturing BVP, EDA, skin temperature, respiration, ECG, and EMG. Each timestep also includes a CoVAS (Continuous Visual Analog Scale) value, a real-time pain rating between 0 and 100 reported by participants, which serves as the ground truth label for our models.

Seconds	Bvp	Eda_E4	Tmp	Resp	Eda_RB	Ecg	Emg	COVAS
2.288	-63.1816	0.4355261	32.59	0.119018555	7.188796997	-0.01606750	-0.01634216	0.25
2.292	-63.1463	0.4355261	32.59	-0.07095336	7.187461853	-0.01059722	0.001739502	0.5
2.296	-63.1110	0.4355261	32.59	0.107828776	7.190831502	-0.01686096	-0.00611877	0.75
2.348	-34.3806	0.4355261	32.59	0.033569336	7.191848755	0.001888275	0.002105713	1.5
2.352	-30.8371	0.4355261	32.59	0.151062012	7.191848755	-0.00283813	0.001441956	2

Figure 1: Example PainMonit Dataset (Experimental)

3.1. Preprocessing

Prior to segmentation and feature extraction, it is necessary to preprocess the physiological signals to reduce noise and artifacts that may affect the performance of the machine learning models. The PainMonit dataset provides synchronized multimodal recordings resampled to a common sampling frequency of 250 Hz, which ensures temporal alignment across all signals (Gouverneur et al., 2024). Filtering techniques are applied to reduce noise components characteristic of each physiological signal. ECG and BVP signals require bandpass filtering to attenuate baseline drift and high frequency noise, while EDA and respiration signals are smoothed using low pass filtering to reduce instantaneous fluctuations and motion artifacts. EMG signals are also filtered to remove movement artifacts and high-frequency noise.

3.2. Segmentation and selection

Since the experimental database contains biomedical signals of participants exposed to 10 second stimulus, it is reasonable to assume that dividing each recording into 10 second windows will most likely capture the most information related to the physiological response (Gouverneur et al., 2024). It is worth noting that using overlapping windows with shorter duration may be more practical for a wearable device implementation since it would allow for more frequent updates. However, this approach may also increase computational cost and potentially reduce prediction accuracy due to the smaller amount of information in each window. Additionally, this model can only be trained with signals that can realistically be acquired by wearable devices, such as BVP, EDA, temperature, and potentially ECG.

3.3. Feature extraction

After segmentation, features are extracted from each window to transform the preprocessed physiological signals into compact representations suitable for machine learning models.

For ECG and BVP signals, heart rate and HRV are the primary informative features (Kasaeyan Naeini et al., 2021). Electrodermal activity signals are analyzed to extract features related to skin conductance variations, such as mean skin conductance level or signal variability, which have been widely used for automated pain assessment (Susam et al., 2018). EMG signals provide features related to muscle activation intensity, including RMS and signal energy, which reflect changes in muscle tension during pain responses (Campbell et al., 2019). Respiration signals capture

breathing dynamics through aspects such as respiration rate and signal variability. Lastly, temperature signals provide features related to slow physiological changes over time such as mean temperature and temporal trends.

3.4. Machine Learning Models

The best approach is to select and train regression models based on the characteristics and nature of the data. For example, if features present a linear relationship with CoVAS, a simpler linear regression algorithm can be sufficient, while non-linear relations are better captured by models like Support Vector Machine (SVM) or Ensemble methods. From this last group, Random Forest is one of the most popular models for biomedical applications since it can recognize complex relationships, is robust to noisy data, and performs well with many features. Additionally, neural networks could be considered if there is a need to capture complex patterns in multimodal physiological data.

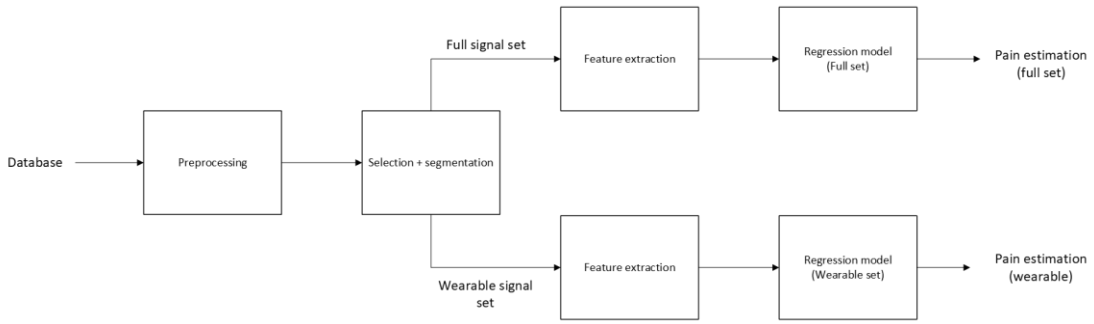

Figure 2: ML Block Diagram

4. Evaluation

In our model, a subject-independent "80-20" splitting scheme will be implemented. We will first set aside 20% of our data set to test our algorithm against at the very end of the process. This set of participants will remain unchanged throughout the entire procedure. The remaining 80% of our set can be called the development set. We will use a 5-Fold Cross Validation scheme where for each fold of the validation process, the participants will not be subjects the model has seen during a given training session. (Hastie et al. 2008)

Our models will be assessed using two metrics suitable for the regression parts of the PainMonit dataset. For the regression model, the MAE will measure the average difference between the predicted pain level and the actual stimulus value. Root Mean Squared Error (RMSE) will be used to measure whether the model makes large errors. RMSE is better at picking outliers than MAE.

The main aim of this project is to measure the "Performance Gap" between clinical-grade sensing and wearable technology. We will be using two different models: the comprehensive model and the wearable-feasible model. The comprehensive model will be trained on the full set of signals given to us through the PainMonit dataset. This data will generally be much more accurate for detection but may be harder to abstract into a wearable device. For wearable devices, we can use the same signals but cutoff our data when the signals are no longer feasible to be measured by a wearable device. This way, we can still utilize the data without compromising feasibility for commercial use. By comparing the MAE and RMSE scores for these two data sets, we can get a general idea of how viable, wearable devices are for continuous clinical pain monitoring.

5. References

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